

Unique spectropolarimetric properties of FRBs codetected by CHIME/FRB and DSA-110

CHIME/FRB AND DSA-110 COLLABORATION MEMBERS¹

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ABSTRACT

We present spectropolarimetric analysis of twelve apparently non-repeating fast radio bursts (FRBs) that are codetected by CHIME/FRB and DSA-110. Jointly leveraging the data from both observatories, we obtain broadband spectral coverage (400 – 1530 MHz) for these codetected FRBs, and constrain subtle propagation effects imprinted in their polarized spectra. We find that the linear polarization fraction in three FRBs decreases as a function of increasing frequency. This spectral signature is commonly observed in pulsars, in which the mixing of two polarized, orthogonal propagation modes causes depolarization at high frequencies; under this framework we interpret our results as evidence for a magnetospheric FRB emission process in these systems. Seven FRBs show evidence of decreasing linear polarization fraction with decreasing frequency, consistent with spectral depolarization due to multipath propagation in the source environment. The prevalence of depolarization in our sample suggests that inhomogeneous environments surrounding FRB sources are more common than previously expected. We also report a $> 3\sigma$ detection of a positive correlation between the extent of depolarization and the amplitude of Faraday rotation measure in depolarizing FRBs, implying that the two arise from the same media. In two FRBs, we measure a decrease in the polarization angle variability with decreasing frequency, as would be expected from amplified scattering of the signal at low frequencies. From this result, we identify a systematic bias towards detecting a disproportionately high fraction of constant PA profiles in current low-frequency (e.g., 400 – 800 MHz) FRB surveys.

Keywords: Radio bursts (1339) — Radio transient sources (2008) — Polarimetry (1278)

1. INTRODUCTION

The physical origins of, and emission mechanisms for, fast radio bursts (FRBs; micro- to millisecond duration, coherent radio transients, mostly from extragalactic distances [D. R. Lorimer et al. 2007](#)) remain elusive. The association of one FRB with a Galactic magnetar (FRB 200428; [CHIME/FRB Collaboration et al. 2020](#); [C. D. Bochenek et al. 2020](#)), and spectropolarimetric properties suggestive of an emission region near the surface of a rapidly rotating compact object (FRB 20221022A; [R. Mckinven et al. 2025](#); [K. Nimmo et al. 2025](#)), potentially point towards FRB emission originating from the magnetospheres of magnetars. However, the specific origins of most FRBs (i.e., over 4500; [CHIME/FRB Collaboration et al. 2026](#)) remain unresolved, and it is not clear whether all FRBs share a common origin or emission mechanism.

Spectropolarimetric properties of FRBs provide key insights into both the FRB emission process and their source environments (e.g., [A. Pandhi et al. 2024](#); [M. B. Sherman et al. 2024](#); [D. R. Scott et al. 2025a](#)). The

intrinsic linear and circular polarization fractions (L/I and $|V|/I$, respectively) of FRB signals can help to constrain the type of emission mechanism responsible for their production (e.g., [Y. Qu & B. Zhang 2023](#)). Temporal variability of the polarization position angle (PA) across the burst duration traces our viewing geometry of the FRB emission region over $\lesssim$ ms timescales (often parameterized as the “rotating vector model” or “RVM” that is applied to radio pulsar emission and to some FRBs, e.g., [V. Radhakrishnan & D. J. Cooke 1969](#); [S. Johnston et al. 2023](#); [R. Mckinven et al. 2025](#)). As FRBs traverse between their emitting source and Earth, their signals accrue a number of frequency-dependent propagation effects from intersecting plasma. Dispersion of radio signals by electrons along the line of sight (LOS), characterized by the dispersion measure (DM), and Faraday rotation of linearly polarized sources, or rotation measure (RM), are two commonly observed properties that inform the density and magnetic field strength of FRB environments. Spectral depolarization has also been observed in a handful of (mostly repeating)

FRBs (Y. Feng et al. 2022; P. A. Uttarkar et al. 2025; A. Pandhi et al. 2024, 2026), and it has been attributed to multipath propagation through an inhomogeneous magnetized plasma.

There are some additional frequency-dependent polarimetric effects that have been studied in radio pulsars that have rarely, if ever, been reported for FRBs. Observations of some pulsars at $\lesssim 1$ GHz, where scatter broadening of the pulse becomes prominent, show that their PA profiles become flattened due to averaging of PAs from different pulse phases within the scattering tail (e.g., M. M. Komesaroff et al. 1972; X. H. Li & J. L. Han 2003; A. Karastergiou 2009; A. Noutsos et al. 2009; P. F. Wang et al. 2023). This effect has been observed for one uniquely bright FRB at 400 – 800 MHz, FRB 20250316A, where coherent PA variability of amplitude $\sim 30^\circ$ at ~ 800 MHz was largely suppressed by scattering at ~ 400 MHz (CHIME/FRB Collaboration et al. 2025).

A number of studies have observed a decreasing linear polarization fraction with increasing observing frequency in pulsar spectra (e.g., R. N. Manchester et al. 1973; K. M. Xilouris et al. 1996; M. M. McKinnon 1997; S. Johnston et al. 2008; A. Noutsos et al. 2015). While there is currently no consensus on the specific mechanism responsible for this phenomenon, most of the proposed models rely on the birefringence of plasma in an extreme magnetized environment, such as the magnetosphere of a neutron star (M. A. Ruderman & P. G. Sutherland 1975; J. J. Barnard & J. Arons 1986; M. M. McKinnon 1997; A. von Hoensbroech et al. 1998). Other explanations of decreasing linear polarization with increasing frequency in pulsars include superimposed orthogonal polarization modes with different spectral indices (A. Karastergiou et al. 2005) and incoherent summing of short duration bursts originating near the neutron star surface (S. Johnston et al. 2008)—both of which still necessitate a magnetospheric emission process. While one FRB shows a potential increase in linear polarization fraction with decreasing frequency between 400 – 800 MHz (A. Pandhi et al. 2024), there has been no definitive detection of this effect in FRBs.

Our capacity to measure many of these spectropolarimetric effects depends on both our observing frequency (ν) and bandwidth ($\Delta\nu$). While most current FRB experiments have sufficient frequency coverage to reliably measure DM and RM for most detected bursts, constraining more subtle spectropolarimetric effects, such as those mentioned above, requires low-frequency and wide-band observations that provide a large lever arm in frequency space. The problem is exasperated as most FRB sources do not repeat

(CHIME/FRB Collaboration et al. 2023), thus necessitating instantaneous broadband observations for a single burst. Uniquely, two of the FRB experiments in the Northern hemisphere, the Canadian Hydrogen Intensity Mapping Experiment FRB Project (CHIME/FRB, observing at 400 – 800 MHz; CHIME/FRB Collaboration et al. 2018) and the Deep Synoptic Array 110 (DSA-110, observing at 1280 – 1530 MHz; C. J. Law et al. 2024), overlap significantly in survey footprint and can codetect FRB sources.

In a companion paper (J. Faber et. al. in prep.; hereafter “paper I”), we introduce a sample of twelve apparently non-repeating FRBs codetected by CHIME/FRB and DSA-110. Here, we leverage the combined bandwidth of the two instruments, together spanning two octaves (400 – 1530 MHz), to conduct a joint spectropolarimetric analysis of this codetected FRB sample. In Section 2, we describe our CHIME/FRB and DSA-110 observations and summarize the process of identifying codetected FRBs. We detail our modeling of the L/I and PA profile as a function of frequency in Section 4. In Section 5, we describe our resulting model fits to the codetected FRB data, finding complex L/I spectra for 8 FRBs, and PA profiles that flatten at low frequencies for two sources. We interpret our measurements in Section 6, showcasing evidence that some of the FRBs may be emitted in the magnetosphere of neutron stars, and we also discuss the implications of low-frequency flattening in PA profiles as an observational systematic in broader polarimetric population studies of FRBs. We summarize our findings and conclude in Section 7.

2. OBSERVATIONS

2.1. CHIME/FRB

CHIME operates between 400 – 800 MHz and, as a transit telescope with no moving parts, it observes the sky above a Declination of -10 degrees. The telescope is located at the Dominion Radio Astrophysical Observatory in British Columbia, Canada. The CHIME/FRB system conducts a real-time FRB search on the CHIME data stream (for details, see CHIME/FRB Collaboration et al. 2018) and records two modes of data for FRB candidates: (i) total intensity data at a time resolution of 0.983 ms and frequency resolution of 24.4 kHz (for FRBs with a signal-to-noise $S/N > 8$) and (ii) full polarization channelized voltage data, or “baseband data”, at a frequency resolution of 390.625 kHz and a time resolution of $2.56 \mu\text{s}$ (for $S/N > 12$).² The median sky posi-

² The threshold is reduced to $S/N > 10$ for subsequent bursts from a known repeating FRB source.

tion uncertainty derived from the total intensity data is relatively coarse, $\sim 0.2^\circ$, with localization confidence intervals often forming non-Gaussian distributions (for details, see CHIME/FRB Collaboration et al. 2019, 2021, 2026). However, raw voltage information is preserved for FRBs that exceed the S/N threshold for saving baseband data, allowing for their localizations to be later refined by a S/N-maximizing search over a dense grid of formed beams; the median sky position uncertainty for FRBs with baseband data is $\sim 20''$ (CHIME/FRB Collaboration et al. 2024). FRBs discovered by CHIME/FRB are broadcast to the community using a low-latency virtual observatory events (VOEvents) system (T. C. Abbott et al. 2025), which contains burst properties and detection information, for example: time of arrival (TOA; topocentric at CHIME and referenced to 400 MHz), real-time detection S/N,³ sky position, and DM.

To derive polarization data products at 400–800 MHz for this work, we employ the CHIME/FRB polarization pipeline (for details, see R. Mckinven et al. 2021; A. Pandhi et al. 2024). The FRB burst envelope temporal limits are identified as the region in which the total intensity is $> 20\%$ of the peak burst intensity. Stokes I , Q , U , and V spectra are produced by integrating over this burst envelope with uniform weights in time. We apply RM-synthesis (M. A. Brentjens & A. G. de Bruyn 2005) to the FRB Stokes spectra, obtaining a robust RM detection if the peak polarized intensity in Faraday depth space is $> 6\sigma$ after applying RM-CLEAN (G. Heald et al. 2009). We then apply the QU-fitting algorithm (implemented with a modified version of V1.2 of the RM-tools package; C. R. Purcell et al. 2020; C. L. Van Eck et al. 2026), using the default model described by R. Mckinven et al. (2021). This QU-fitting step provides an independent measurement of the FRB RM⁴, as well as an estimate of the physical (cable) delay between the two linear polarizations, which can act as a source of contamination between Stokes U and V . We then return to the Stokes dynamic spectra, derotate them given the FRB RM, apply a correction based on the measured cable delay term, and integrate over the burst envelope temporal limits to obtain a set of “corrected” Stokes I , Q , U , and V spectra that we use for any further analysis.

³ Details of the real-time CHIME/FRB search pipeline are described by (CHIME/FRB Collaboration et al. 2018).

⁴ In some cases, instrumental polarization can cause the sign of the RM derived via QU-fitting to be incorrect (see Appendix A by R. Mckinven et al. 2021). Thus, we opt to use the RM estimate from RM-synthesis throughout this work. A detailed analysis to mitigate this QU-fitting sign error will be presented in upcoming work by M. Ng et al. (in prep.).

For any bursts that do not satisfy the 6σ linearly polarized signal-to-noise criterion, we only measure an upper limit on their L/I averaged across 400–800 MHz of $6/(S/N(I))$, where $S/N(I)$ is the signal-to-noise of the Stokes I signal (A. Pandhi et al. 2024).

2.2. DSA-110

The DSA-110 is a radio interferometer located at the Owens Valley Radio Observatory that is designed to detect and precisely localize (to a radius of $\sim$ a few arcseconds) FRBs between 1280 – 1530 MHz. The data presented in this work were collected during the science commissioning stage in which the interferometer was comprised of sixty-three, 4.65 m, antennas, each equipped with dual-linear-polarization receivers that are distributed across a 2.5 km area (C. J. Law et al. 2024). During commissioning the antenna were pointed at a declination of 71.6° with an effective field of view of 10° . An FRB search was conducted at a frequency and time resolution of 30.52 kHz and 32.768 μ s, respectively, using 48 core antennas while the remaining 15 antennas, with longer baselines, were used to assist in FRB localization (M. B. Sherman et al. 2024). **Following a real-time detection, candidate positions from the coherent search beams are refined offline by forming interferometric visibilities from the saved voltage data of the core and outrigger antennas, calibrating them using nearby calibrator observations and an NRAO VLA Sky Survey (NVSS) sky model, and imaging the burst with the long-baseline data (V. Ravi et al. 2023). The sky position is measured by fitting a two-dimensional Gaussian to the deconvolved burst image, yielding typical 90% confidence uncertainties of $\lesssim 2''$ (C. J. Law et al. 2024; V. Ravi et al. 2023).** Full-polarization channelized voltage data are saved for FRB candidates that pass the real-time search trigger (C. J. Law et al. 2024), namely: (i) **a detection $S/N > 8.5$ in any coherent search beam,** (ii) **no association with strong terrestrial radio-frequency interference,** and (iii) **$DM > 50 \text{ pc cm}^{-3}$ and $DM > 0.75 DM_{MW}$ along the line of sight, where DM_{MW} is the Galactic contribution from J. M. Cordes & T. J. W. Lazio (2002). For detailed descriptions of the DSA-110 system and FRB search during science commissioning, see (V. Ravi et al. 2023; C. J. Law et al. 2024).**

Here, we summarize the polarization pipeline, described in detail by M. B. Sherman et al. (2024), that we use to derive polarization data products at 1280 – 1530 MHz for this work. First, calibration is applied to reduce the affect of instrumental polarization due to

differences in gain (leakage between Stokes I and Q) and phase (leakage between Stokes U and V ; also called “cable delay” in Section 2.1) between the two linearly polarized receivers. Two bright radio sources, one with known polarization (3C286; $L/I = 0.095$ at 1400 MHz) and one unpolarized (3C48; $L/I \lesssim 0.005$), are used to derive a Jones matrix and correct for these instrumental polarization effects. Absolute PAs are estimated by treating the DSA-110 as a single antenna, deriving its elevation and azimuth given the position of the FRB-detecting beam and the detection local sidereal time to estimate the parallactic angle, which is then applied as a correction to the observed PA (for further details regarding the polarization calibration, see Appendices D–G by M. B. Sherman et al. 2024).

For a given FRB, Stokes spectra are generated by averaging over the temporal extent of the burst, which are determined manually to encompass all components in the case of multi-component bursts. Similar to the CHIME/FRB polarization pipeline, an RM search is conducted on these Stokes spectra using the version of RM-synthesis implemented in the `RM-tools` package (M. A. Brentjens & A. G. de Bruyn 2005; C. R. Purcell et al. 2020; C. L. Van Eck et al. 2026). Instead of searching for an RM across the entire burst envelope, the linearly polarized signal in the Faraday dispersion function (FDF) is maximized for each time step in the burst, and the resulting time-resolved RMs across the burst are averaged to get an average RM for the burst. If a significant (i.e., linearly-polarized signal to noise exceeding 9σ) is detected, the Stokes dynamic spectra are derotated to correct for the measured RM. Integrating these derotated Stokes spectra across the temporal extent of the FRB then provides the equivalent set of “corrected” Stokes I , Q , U , and V spectra at 1280 – 1530 MHz that we use for further analysis.

3. SEARCH FOR CODETECTED FRBS

We identify twelve apparently non-repeating FRBs jointly detected by CHIME/FRB and DSA-110 between 2022 February and 2024 February. Candidate pairs are required to have consistent sky positions and arrival times: after correcting for the inter-site geometric light-travel delay and cold-plasma dispersion to a common reference frequency of 400 MHz, the timing residuals across the sample are consistent with the combined instrumental uncertainties (median ~ 2 ms; paper I). Angular separations between the CHIME/FRB and DSA-110 localizations are $\lesssim$ a few arcminutes for every burst. Independent dispersion measures estimated at

each facility agree where both are well constrained. Each association is additionally vetted by visual comparison of the total-intensity dynamic spectra, including agreement in the number of temporal components across the two bands; the clearest multi-component example is FRB 20220310F, for which the detailed morphology comparison is presented in paper I. The resulting sample comprises twelve non-repeating, broadband events spanning 262–960 pc cm^{−3} in DM; spectroscopic host-galaxy redshifts are available for nine bursts, while FRBs 20230325A, 20230814B, and 20240122A lack a confirmed host redshift. Scattering timescales, host-galaxy associations, and the full cross-matching methodology are reported in paper I.

4. SPECTROPOLARIMETRIC MODELING

We focus on analyzing the L/I and PA profile of the codetected FRBs as a function of frequency, and we quantify and model their spectral properties based on frequency-dependent polarization effects that have been previously observed in other FRBs and in some radio pulsars. In this work, we do not analyze the circular polarization fraction, $|V|/I$, as CHIME/FRB data suffers from unmodeled instrumental sources of circular polarization that can lead to $|V|/I \sim 20\%$ uncertainties and limit our sensitivity to frequency-dependent circular polarization effects (R. Mckinven et al. 2021).

4.1. Linear polarization

We model the observed linear polarization fraction as a function of frequency, $L/I(\nu)$, using three different frameworks, each adding a layer of complexity to describe potential propagation effects near the FRB engine and in its source environment. These models are described below and example $L/I(\nu)$ spectra between 400 – 1530 MHz for each of the three scenarios, using some fiducial model parameters, are presented in Figure 1.

4.1.1. Model 1: Constant L/I

In the most straightforward model, the FRB is emitted at a constant intrinsic linear polarization fraction, $(L/I)_{\text{int}}$, across its emitting bandwidth, and the signal is unaffected by any propagation effects. Then, the observed $L/I(\nu)$ is simply:

$$L/I(\nu) = (L/I)_{\text{int}}. \quad (1)$$

Examples of $L/I(\nu)$ following Equation 1 for a range of fiducial values are presented in the left panel of Figure 1.

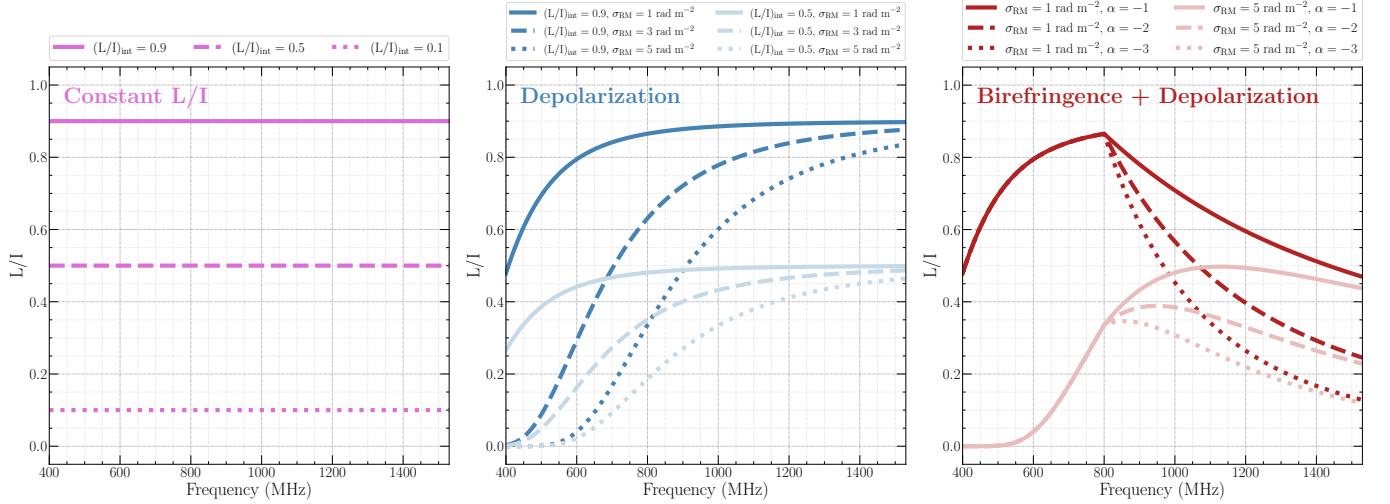


Figure 1. (Left) Toy model of a constant L/I versus frequency for different values of the intrinsic linear polarization fraction, $(L/I)_{\text{int}}$. (Middle) Toy model of depolarization following Equation 2 for various input σ_{RM} values with $(L/I)_{\text{int}} = 0.9$ (dark lines) and $(L/I)_{\text{int}} = 0.5$ (faint lines). (Right) Toy model combining birefringence and depolarization effects (see Equation 3). Dark lines show the model behavior for various α indices with $\sigma_{\text{RM}} = 1 \text{ rad m}^{-2}$; faint lines show curves with $\sigma_{\text{RM}} = 5 \text{ rad m}^{-2}$ and the same range of α . All shown curves for the birefringence and depolarization model assume $(L/I)_{\text{int}} = 0.9$ and $\nu_c = 800 \text{ MHz}$. The frequency range in all panels spans 400 MHz (bottom of the CHIME/FRB band) to 1530 MHz (top of the DSA-110 band).

4.1.2. Model 2: Depolarization

Now, the FRB emitted with $(L/I)_{\text{int}}$ passes through inhomogeneous plasma, causing the signal to be scattered to a finite angular size. It then undergoes differential Faraday rotation due to intervening spatial fluctuations in electron density and/or magnetic field strength. When the FRB is detected, these spatial RM variations are averaged across the observing beam, leading to a decrease in the observed linear polarization, $L/I(\nu)$, relative to $(L/I)_{\text{int}}$ (referenced to zero wavelength):

$$L/I(\nu) = (L/I)_{\text{int}} \exp(-2c^4 \nu^{-4} \sigma_{\text{RM}}^2). \quad (2)$$

Here, c is the speed of light and σ_{RM} is a measure of the inhomogeneity in the scattering medium (B. J. Burn 1966). The spectral depolarization framework in Equation 2 has been adopted by previous FRB studies (e.g., Y. Feng et al. 2022; P. A. Uttarkar et al. 2025; A. Pandhi et al. 2026), allowing us to directly compare our results to the literature. The middle panel of Figure 1 shows expected $L/I(\nu)$ spectra for a range of fiducial σ_{RM} and $(L/I)_{\text{int}}$ values.

4.1.3. Model 3: Birefringence and depolarization

Evidence for decreasing linear polarization fraction with increasing frequency has been observed in many radio pulsar spectra across several octaves in frequency (e.g., D. Morris et al. 1981; D. M. Gould & A. G. Lyne 1998; S. Johnston et al. 2008; A. Noutsos et al. 2015).

Radio emission from pulsars is thought to originate in the magnetosphere of the neutron star above their polar

caps, with the emitting frequency relating to the height above the neutron star surface the emission is generated (sometimes referred to as “radius-to-frequency mapping”; M. A. Ruderman & P. G. Sutherland 1975). Their polarized emission is thought to be comprised of two orthogonal propagation modes, the ordinary and extraordinary modes (R. N. Manchester et al. 1975). The differing, frequency-dependent refractive indices of these two modes cause their beaming angles to diverge as they propagate through a birefringent medium, such as a neutron star magnetosphere (J. J. Barnard & J. Arons 1986). The angular separation between the two modes increases at lower frequencies (higher emission heights) where refraction is stronger such that the observer may only see emission from one of the two modes, which are each highly polarized. At higher frequencies (lower emission heights), the beaming angle between the two modes decreases and the observed emission becomes depolarized from the mixing of the orthogonal polarization modes. Empirically, R. N. Manchester et al. (1973) describe the $L/I(\nu)$ of these radio pulsars as a power-law decrease above a characteristic frequency, ν_c , where the orthogonal modes become increasingly mixed, and the $L/I(\nu)$ is constant below ν_c , when the observed emission is dominated by only one mode.

If FRB emission is produced in the magnetosphere of a highly magnetized neutron star, it is plausible that the FRB spectra may have signatures of depolarization at both high (due to the birefringent magnetosphere) and low (due to propagation through an inhomogeneous

magnetoionic plasma in its local environment) frequencies. We combine these two effects in Model 3, such that the observed linear polarization fraction of the FRB emission is:

$$L/I(\nu) = \exp(-2c^4\nu^{-4}\sigma_{\text{RM}}^2) \times \begin{cases} (L/I)_{\text{int}}, & \nu \leq \nu_c \\ (L/I)_{\text{int}} \left(\frac{\nu}{\nu_c}\right)^\alpha, & \nu > \nu_c \end{cases}. \quad (3)$$

Example $L/I(\nu)$ spectra for fiducial values of $(L/I)_{\text{int}}$, ν_c , σ_{RM} , and α are plotted in the rightmost panel of Figure 1.

4.2. Polarization angle

From the derotated Stokes Q and U dynamic spectra (denoted by the subscript “derot” for clarity) we can derive the observed PA as a function of time, t , across the burst envelope as:

$$\text{PA}(t) = \frac{1}{2} \arctan \left(\frac{U_{\text{derot}}(t)}{Q_{\text{derot}}(t)} \right). \quad (4)$$

PA uncertainties are calculated by propagating uncertainties in the Q_{derot} and U_{derot} spectra, following the procedure detailed by T. Vernstrom et al. (2019). We mask out any points in the PA profile below a linearly-polarized $\text{S/N} < 5$, a conservative threshold above which the PA uncertainties have shown to be robust (see Appendix B by A. Pandhi et al. 2024). Since the CHIME/FRB polarization pipeline does not calibrate for an absolute PA due to unknown beam phase effects off of CHIME’s meridian axis, we focus our analysis on the relative PA evolution across the burst. Hence, we center both the 400 – 800 MHz and 1280 – 1530 MHz PA profiles to have a median of 0° . Following the procedure described by A. Pandhi et al. (2024), we quantify the magnitude of PA variability in a given $\text{PA}(t)$ timeseries by evaluating the goodness of fit between the observed PA profile, centered at 0° , and a constant $\text{PA}(t) = 0^\circ$ model using a reduced χ_ν^2 test. We a threshold of $\chi_\nu^2 \geq 5$, above which we consider the PA profile to be variable.

Given our large effective bandwidth for these code-detected FRBs, we aim to characterize whether PA profiles of scattered FRBs become flattened at low frequencies due to mixing of signal from different pulse phases in the scattering tail. In the FRB literature, this effect is most clearly seen in FRB 20250316A, in which a clear PA swing of 30° at ~ 800 MHz becomes suppressed by scatter broadening at ~ 400 MHz, where the PA is almost entirely constant across the burst envelope (see Figure 5 by CHIME/FRB Collaboration et al. 2025). We search

for such signals in our code-detected FRBs by applying the following criteria: (i) $\chi_\nu^2 \geq 5$ at 1280 – 1530 MHz while $\chi_\nu^2 < 5$ at 400 – 800 MHz, (ii) the amplitude of the maximum PA variation across the profile decreases from 1280 – 1530 to 400 – 800 MHz and (iii) there is a clear coherent PA swing at 1280 – 1530 MHz. The latter criterion ensures that the large χ_ν^2 is not dominated by, e.g., PA variation between subcomponents, each of which may have constant PA profiles. In cases where the FRB is sufficiently bright at 400 – 800 MHz, we further subband the data to compute PA profiles at 400 – 600 MHz and 600 – 800 MHz, respectively.

5. RESULTS

5.1. FRB magnetoionic environments

The observed RMs are summarized the results in Table 1. Performing RM-synthesis on joint CHIME/FRB–DSA-110 spectra only marginally improves the theoretical RM uncertainty while causing significant artifacts in the FDF due to differences in the relative polarized intensity scalings between instruments, and irregular frequency sampling. Thus, we opt to report the RM_{obs} from the CHIME/FRB data as the lower frequency and larger bandwidth observations provide greater accuracy in the RM determination; we confirm that the FDF peaks of the CHIME/FRB and DSA-110 data agree to within their respective full-width-half-maximum for each code-detected FRB. Polarization properties of these FRBs will also appear in a future CHIME/FRB polarization catalog (M. Ng et al. in prep.) and in M. B. Sherman et al. (2024). The observed DM, DM_{obs} , and host galaxy redshift, z , of these FRBs are reported in paper I and adopted for this work.

DM_{obs} and RM_{obs} can be written as the sum of individual contributions from plasma along the LOS:

$$\text{DM}_{\text{obs}} = \text{DM}_{\text{ion}} + \text{DM}_{\text{disk}} + \text{DM}_{\text{halo}} + \text{DM}_{\text{IGM}}(z) + \frac{\text{DM}_{\text{host}}}{(1+z)} \text{ pc cm}^{-3}; \quad (5)$$

$$\text{RM}_{\text{obs}} = \text{RM}_{\text{ion}} + \text{RM}_{\text{disk}} + \text{RM}_{\text{halo}} + \text{RM}_{\text{IGM}}(z) + \frac{\text{RM}_{\text{host}}}{(1+z)^2} \text{ rad m}^{-2}, \quad (6)$$

where “ion” refers to the Earth’s ionosphere, “disk” and “halo”, are the Milky Way disk and halo, respectively, “IGM” is the intergalactic medium and “host” is the host galaxy. We can calculate the electron density-weighted average LOS magnetic field strength in the FRB host galaxy using the DM_{host} and RM_{host} :

$$\langle B_{\parallel, \text{host}} \rangle = 1.232 \frac{\text{RM}_{\text{host}}}{\text{DM}_{\text{host}}} \mu\text{G}. \quad (7)$$

We employ Monte Carlo random sampling to determine DM_{host} , RM_{host} , and $\langle B_{\parallel, \text{host}} \rangle$, following the procedure described by A. Pandhi et al. (2025). For each FRB, we perform 10^4 trials, drawing foreground DM and RM contributions from assumed distributions and using the host galaxy redshift to convert to rest-frame quantities. The median of the resulting DM_{host} , RM_{host} , and $\langle B_{\parallel, \text{host}} \rangle$ distributions are taken as the maximum likelihood estimates with 68% confidence intervals representing the associated uncertainties. DM_{obs} and RM_{obs} are drawn from Gaussian distributions centered on their measured values and with standard deviation equal to their uncertainties. We assume that the ionospheric contribution is negligible (typically $\text{DM}_{\text{ion}} \lesssim 10^{-5} \text{ pc cm}^{-3}$ and $\text{RM}_{\text{ion}} \lesssim 1 \text{ rad m}^{-2}$; M. T. Lam et al. 2016; C. Sobey et al. 2019). DM_{disk} is drawn from a Gaussian with mean and standard deviation equal to the DM and uncertainty estimated by the NE2025 model towards the FRB position (S. K. Ocker & J. M. Cordes 2026). DM_{halo} is drawn from a Lognormal distribution with mean $\log_{10}(30 \text{ pc cm}^{-3})$ and a 0.2 dex standard deviation (K. Dolag et al. 2015; S. Yamasaki & T. Totani 2020; A. M. Cook et al. 2023). $\text{RM}_{\text{disk}} + \text{RM}_{\text{halo}}$ follow a Gaussian distribution based on the reconstructed Galactic mean RM and variance at the FRB position (S. Hutschenreuter et al. 2022). DM_{IGM} is sampled from the probability distribution function presented in Equation 1 of J. Baptista et al. (2024), based on the empirical $\text{DM}_{\text{IGM}} - z$ relation. RM_{IGM} follows a Gaussian distribution centered at 0 rad m^{-2} with standard deviation 6 rad m^{-2} (D. H. F. M. Schnitzeler 2010). The best-fit DM_{host} , RM_{host} , and $\langle B_{\parallel, \text{host}} \rangle$ for each codetected FRB associated with a host galaxy with a measured spectroscopic redshift is presented in Table 1.

5.2. Linear polarization

We utilize an ensemble Markov Chain Monte Carlo (MCMC) sampling routine, implemented with the `emcee` package (D. Foreman-Mackey et al. 2013), to fit the observed L/I spectra of each codetected FRB to the three models presented in Section 4.1 (Equations 1–3). Assuming Gaussian uncertainties, the log-likelihood is defined as:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_i \left[\left(\frac{(L/I)_i - (L/I)_{\text{model},i}}{\sigma_{(L/I),i}} \right)^2 + \ln \left(2\pi\sigma_{(L/I),i}^2 \right) \right], \quad (8)$$

where $(L/I)_i$ is the observed linear polarization fraction and $\sigma_{(L/I),i}$ is the associated measurement uncertainty; $(L/I)_{\text{model},i}$ is the expected model L/I following one

Table 1. Summary of magnetoionic properties for the 12 codetected FRB host environments.

TNS Name	RM_{obs} (rad m^{-2})	RM_{host} (rad m^{-2})	DM_{host} (pc cm^{-3})	$\langle B_{\parallel, \text{host}} \rangle$ (μG)
FRB 20220207C	$-161.54(1)$	-170^{+20}_{-19}	141^{+18}_{-27}	$-1.5^{+0.2}_{-0.4}$
FRB 20220310F	$-5.3(2)$	$+19^{+21}_{-21}$	136^{+69}_{-81}	$+0.18^{+0.36}_{-0.19}$
FRB 20220506D	$+32(4)^a$	$+70^{+27}_{-28}$	133^{+50}_{-70}	$+0.67^{+0.73}_{-0.30}$
FRB 20221113A	$-13.65(9)$	$+50^{+30}_{-30}$	172^{+48}_{-76}	$+0.38^{+0.37}_{-0.23}$
FRB 20221203A	$+21.0(2)$	$+59^{+30}_{-31}$	242^{+89}_{-126}	$+0.31^{+0.36}_{-0.17}$
FRB 20230307A	$-473.49(9)$	-756^{+15}_{-15}	464^{+56}_{-130}	$-2.0^{+0.2}_{-0.8}$
FRB 20230325A ^b	$-10.48(2)$	—	—	—
FRB 20230814B ^b	$+23.3(1)$	—	—	—
FRB 20230913A	$-81.39(3)$	-164^{+27}_{-25}	285^{+59}_{-111}	$-0.72^{+0.17}_{-0.45}$
FRB 20240122A^b	$-2.4(3)$	—	—	—
FRB 20240203A	$+178.18(9)$	$+221^{+19}_{-19}$	135^{+22}_{-35}	$+2.0^{+0.7}_{-0.3}$
FRB 20240229A	$-11.21(1)$	$+3^{+13}_{-13}$	305^{+57}_{-113}	$+0.01^{+0.06}_{-0.06}$

^aObtained from DSA-110 data as the linearly polarized intensity of the CHIME/FRB detection was too low to robustly measure RM.

^bDoes not have an associated host galaxy with a measured spectroscopic redshift.

of Equations 1, 2, or 3. We adopt uniform priors over the following ranges: $0 \leq (L/I)_{\text{int}} \leq 1$, $0 \text{ rad m}^{-2} \leq \sigma_{\text{RM}} \leq 100 \text{ rad m}^{-2}$, $0 \text{ MHz} < \nu_c < 10^4 \text{ MHz}$, and $-10 < \alpha < 0$. Maximum-likelihood estimates for each parameter are determined as the medians of the resulting posterior distributions, with uncertainties quoted as the 16th and 84th percentiles. We conduct model comparison in two steps: (i) we compute the Bayesian Information Criterion (BIC; G. Schwarz 1978) and determine which model minimizes the BIC for a given FRB L/I spectra, and (ii) if the BIC between two or more models are similar (to within $\lesssim 1\%$) and their model spectra are visually similar, we default to the simplest (i.e., with the least physical assumptions) model.

For each codetected FRB, we plot the observed L/I spectra and the MCMC model fits for our three $L/I(\nu)$ models in Figure 2. The best-fit L/I model is identified in the bottom right of each panel and the maximum-likelihood estimates for those model parameters are provided in Table 2. Note that the lowest frequency bin for both FRBs 20220506D and 20221203A are upper limits in L/I but, for the purposes of our model fitting, they are treated as data points. This is specifically because each of the upper limits show a large downturn in L/I from the next-highest frequency bin. By treating the upper limit as a measured L/I , we are effectively able to constrain an upper limit on σ_{RM} in these two cases

rather than make a direct measurement (this caveat is also reflected in Table 2).

Overall, we find that 4/12 FRBs favor the constant L/I model, 5/12 prefer the depolarization model, and 3/12 are best described by the combination of magnetospheric birefringence effects and beam depolarization. Two of the three FRBs in the latter category (FRBs 20230307A and 20240229A) show a clear downtick in L/I towards 400 MHz, while the L/I of FRB 20230913A increases from DSA-110 to CHIME/FRB frequencies, and it remains approximately constant across 400 – 800 MHz. Thus, significant depolarization is observed in 7/12 FRBs in our sample.

5.3. Polarization angle

We find that two FRBs in our sample (FRBs 20220310F and 20230814B) have variable PA profiles as a function of frequency; both of their PA profiles, integrated over multiple frequency ranges between 400 – 1530 MHz, are presented in Figure 3. FRB 20220310F shows a $\sim 70^\circ$ PA swing over ~ 0.1 ms at 1280 – 1530 MHz, which is reduced to a flat PA profile spanning ~ 1 ms with a maximum PA variation of only $\sim 30^\circ$. At 1280 – 1530 MHz, FRB 20230814A shows a $\sim 50^\circ$ PA swing over ~ 0.1 ms. Dividing the CHIME/FRB band in two, we find that the maximum PA variation is reduced to $\sim 20^\circ$ at 600 – 800 MHz, and further to $\sim 10^\circ$ at 400 – 600 MHz. Most of the remaining 10/12 FRBs in our sample show flat PA profiles at both CHIME/FRB and DSA-110 frequencies, and a few have such a low S/N in one of the two datasets that we cannot construct a robust PA profile over multiple frequency bands.

Note that the same limitations discussed at the end of Section 5.2 also apply here since, in theory, two subcomponents of the same burst may have different temporal PA behavior. However, we only observe flattening of the PA towards lower frequencies or flat PA profiles at both frequency band, and not PA profiles with significant variability only at 400 – 800 MHz. If the frequency-dependent structure in the PA profiles originated from the observation of different emitting components, there should not be a preference for flat PAs to be at lower frequencies versus higher frequencies. Therefore, we argue that the spectral evolution of PA profiles in our code-detected FRBs is not dominated by intraburst variation of polarization properties.

5.4. Correlation between depolarization, Faraday rotation, and scattering

Correlations between σ_{RM} , $|\text{RM}|$, and scattering time scale (τ) have been suggested as signposts for FRBs

originating in dense, magnetized, and inhomogeneous environments. However, the significance of these correlations have only been marginal thus far, chiefly owing to the small sample size of depolarizing FRBs (Y. Feng et al. 2022; P. A. Uttarkar et al. 2025; A. Pandhi et al. 2026). Here, we revisit the $\sigma_{\text{RM}} - |\text{RM}|$ and $\sigma_{\text{RM}} - \tau$ relationships with all depolarizing FRBs in the literature in addition to the four FRBs in this work that show evidence for depolarization, have measured scattering timescales, and are associated with host galaxies with spectroscopic redshifts (FRBs 20220207C, 20221113A, 20230307A, and 20240229A; for measurements of their scattering timescales and host galaxy associations, see paper I).

We follow the steps described by A. Pandhi et al. (2026) to evaluate these correlations, specifically accounting for a conversion to the FRB rest-frame and for RM variability in some repeating FRBs. First, we convert to the rest-frame RM scatter, $\sigma_{\text{RM,host}}$, given the cosmological redshift, z , of the FRB host galaxies (A. R. Taylor et al. 2024),

$$\sigma_{\text{RM,host}} = \sigma_{\text{RM}}(1+z)^2. \quad (9)$$

Following Equation 6, we assume $\text{RM}_{\text{ion}} = \text{RM}_{\text{IGM}} = 0 \text{ rad m}^{-2}$, subtract the Galactic RM as measured by S. Hutschenreuter et al. (2022), convert to the rest frame by applying a $(1+z)^2$ correction factor, and the the absolute value to get $|\text{RM}_{\text{host}}|$. If the source is a repeating FRB, we take the median of $|\text{RM}_{\text{host}}|$ across all polarized bursts.

In the left panel of Figure 4, we plot the $|\text{RM}_{\text{host}}|$ versus $\sigma_{\text{RM,host}}$ of depolarizing FRBs in this work compared to other known depolarizing FRBs (Y. Feng et al. 2022; P. A. Uttarkar et al. 2025; A. Pandhi et al. 2026). For the repeating FRBs, we plot the 68% confidence interval around the median $|\text{RM}_{\text{host}}|$ as shaded regions. We undertake a Pearson R test on the $\log_{10}(\sigma_{\text{RM,host}}) - \log_{10}(|\text{RM}_{\text{host}}|)$ relationship, finding a test statistic equal to 0.772 (positive correlation) with a p-value of 0.002 ($\sim 3.1\sigma$). Thus, we consider this correlation to be statistically significant.

Next, we restrict our depolarizing FRB sample to only those with measured scattering timescales exceeding the contribution predicted by the NE2025 model (scaled to the same reference frequency assuming a frequency dependence of ν^{-4} ; S. K. Ocker & J. M. Cordes 2026) by more than an order of magnitude (D. C. Price et al. 2019; A. E. Lanman et al. 2022; S. K. Ocker et al. 2023; A. P. Curtin et al. 2025; D. R. Scott et al. 2025b). Assuming negligible scattering contributing from any intervening galaxy halos, enforcing this threshold allows us to have a high level of confidence that the observed scat-

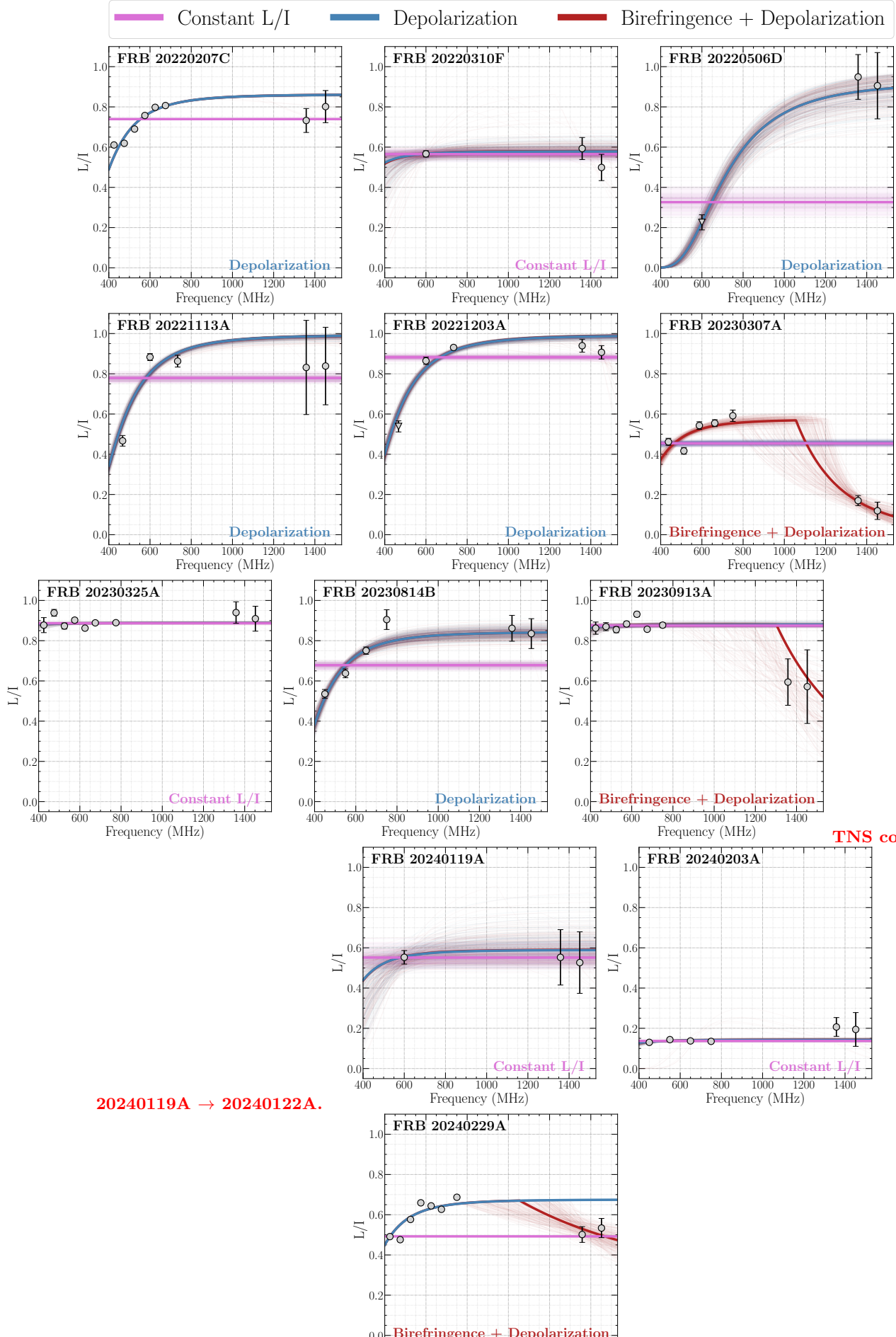


Table 2. Summary of best-fit parameters from L/I spectra modeling.

TNS Name	Best-fit model	$(L/I)_{\text{int}}$	σ_{RM} (rad m ⁻²)	ν_c (MHz)	α
FRB 20220207C	Depolarization	$0.862^{+0.03}_{-0.03}$	$0.946^{+0.008}_{-0.008}$	—	—
FRB 20220310F	Constant L/I	$0.56^{+0.01}_{-0.01}$	—	—	—
FRB 20220506D	Depolarization	$0.92^{+0.05}_{-0.08}$	$> 3.4^{+0.2}_{-0.2}{}^a$	—	—
FRB 20221113A	Depolarization	$0.993^{+0.005}_{-0.010}$	$1.30^{+0.04}_{-0.04}$	—	—
FRB 20221203A	Depolarization	$0.991^{+0.006}_{-0.010}$	$> 1.20^{+0.04}_{-0.05}{}^a$	—	—
FRB 20230307A	Birefringence + Depolarization	$0.57^{+0.02}_{-0.02}$	$0.82^{+0.08}_{-0.08}$	1057^{+108}_{-177}	-5^{+2}_{-3}
FRB 20230325A	Constant L/I	$0.887^{+0.002}_{-0.002}$	—	—	—
FRB 20230814B	Depolarization	$0.844^{+0.02}_{-0.02}$	$1.12^{+0.06}_{-0.07}$	—	—
FRB 20230913A	Birefringence + Depolarization	$0.880^{+0.007}_{-0.006}$	$0.2^{+0.1}_{-0.1}$	1302^{+5530}_{-275}	-3^{+2}_{-4}
FRB 20240122A	Constant L/I	$0.55^{+0.03}_{-0.03}$	—	—	—
FRB 20240203A	Constant L/I	$0.137^{+0.002}_{-0.002}$	—	—	—
FRB 20240229A	Birefringence + Depolarization	$0.676^{+0.002}_{-0.002}$	$0.803^{+0.03}_{-0.03}$	1054^{+150}_{-191}	$-1.0^{+0.4}_{-0.9}$

^aThe lowest frequency bin, in which most of the depolarization occurs, is an upper limit on L/I , thus only a σ_{RM} lower limit is provided by the model fitting.

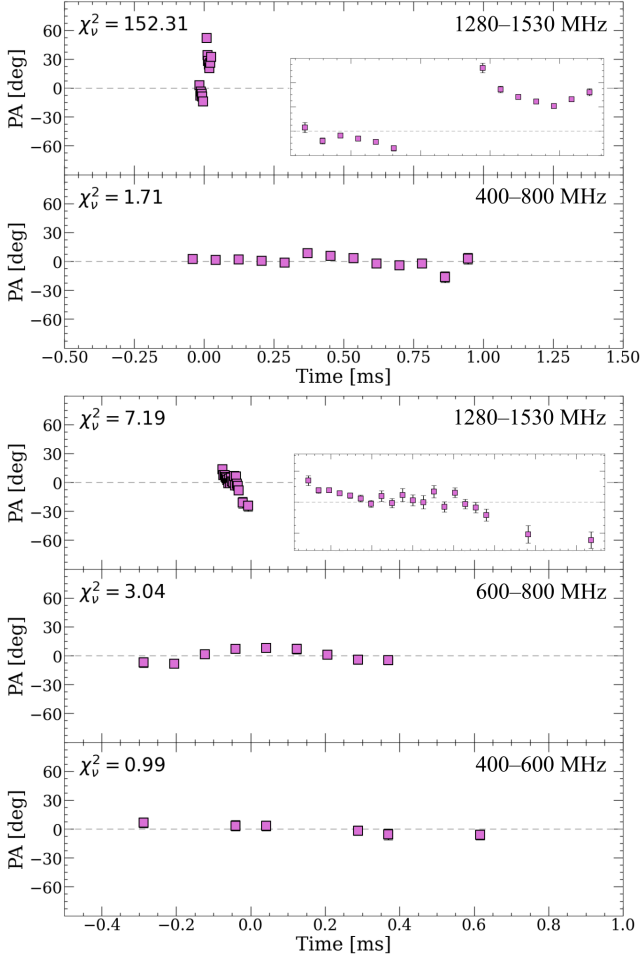


Figure 3. (Top) Temporal PA profile of FRB 20220310F integrated over 1280 – 1530 MHz and 400 – 800 MHz. The reduced chi squared, χ^2_v , from comparing the PA profiles to a constant PA of 0° is presented in the top left of each panel. An inset showing the zoomed in (spanning ~ 0.1 ms) PA profile at the higher-frequency band is overplotted in the first panel. (Bottom) The same plot but for FRB 20230814B. The CHIME/FRB detection of this burst has sufficient linearly polarized intensity to divide the data into two subbands, thus PA profiles integrated over 400–600 MHz and 600–800 MHz are presented.

tering timescales are dominated by contributions from the FRB host galaxy or immediate environment. After this cut, we retain all of the depolarizing FRBs in this work and 7/9 of the depolarizing FRBs in the literature. Converting the scattering timescale to the FRB rest frame to account for cosmological time dilation, assuming $\tau \propto \nu^{-4}$, we apply the equation

$$\tau_{\text{host}} = \tau / (1 + z)^3, \quad (10)$$

(J.-P. Macquart & J. Y. Koay 2013). Note that if the scattering index diverges from the assumed ν^{-4} , it can change the significance of any observed correlation. In

the right panel of Figure 4, we plot τ_{host} versus $\sigma_{\text{RM,host}}$ of depolarizing FRBs. Again, the 68% confidence intervals around the median τ_{host} for repeaters are plotted as shaded regions. Performing a Pearson R test on $\log_{10}(\sigma_{\text{RM,host}}) - \log_{10}(\tau_{\text{host}})$, we find a test statistic of 0.781 with a p-value equal to 0.005 ($\sim 2.8\sigma$), making this correlation marginally significant.

In Figure 5, we plot the cumulative distribution function of τ (in the observer frame since not all FRBs in our sample are associated with host galaxies) for FRBs with (blue) and without (red) depolarization of their linearly polarized spectra. For comparison, we also show the distribution of τ measured in the first CHIME/FRB base-band catalog (K. R. Sand et al. 2025). Here, we have employed statistical methods that take censored data into account (see, e.g., E. D. Feigelson & G. J. Babu 2012), via the NADA V1.6–1.2 package⁵ in R (D. R. Helsel 2005; L. Lopaka 2020; R Core Team 2020), to incorporate the τ upper limits into the empirical cumulative distribution function. While it is difficult to draw definitive conclusions due to our limited sample size, we note that FRBs undergoing depolarization peak at larger scattering timescale than those with no measurable depolarization above 400 MHz. The mean scattering timescale, referenced to 600 MHz, for the seven depolarizing FRBs is $\tau = 2.7 \pm 1.9$ ms, while the mean for FRBs without depolarization is $\tau = 1.4 \pm 0.3$ ms.

6. DISCUSSION

Our unique observing frequency coverage obtained from codetections of FRBs between CHIME/FRB and DSA-110 provides both broadband and low frequency observations that enable the characterization of spectropolarimetric propagation effects that are, thus far, underexplored in FRBs. Namely, we find that only a third (4/12 sources) of our FRB sample exhibits constant L/I as a function of frequency, as would be expected for from FRB emission that is not strongly distorted by propagation effects in the immediate source environment. Instead FRBs commonly show evidence for depolarization towards 400 MHz (7/12 sources), and 3/12 sources show an increase in L/I from the DSA-110 band to the CHIME/FRB band. A suppression of PA variability is also observed in two sources that exhibit clear PA swings at 1400 MHz and they appear to have constant PA profiles at 400 – 800 MHz, where scatter broadening is more prominent. Notably, while these unique propagation effects are seen in a large portion of our FRB sample, these sources do not otherwise appear to be originating from noteworthy local environments

⁵ <https://rdrr.io/cran/NADA/>

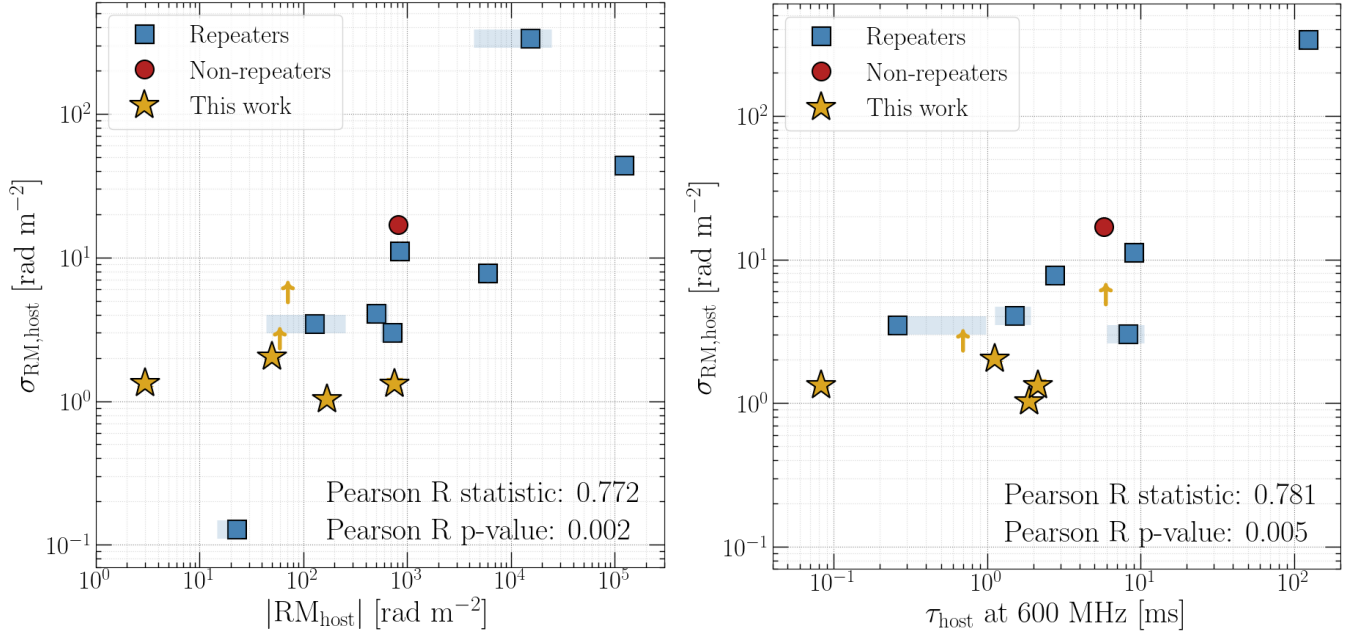


Figure 4. (Left) Rest-frame scatter in the observed RM, $\sigma_{\text{RM,host}}$, plotted as a function of $|\text{RM}_{\text{host}}|$ for the four codetected FRBs that show depolarization signatures and have a host galaxy with a measured spectroscopic redshift (gold stars). The $\sigma_{\text{RM,host}}$ lower limits for FRBs 20220506D and 20221203A are plotted as gold arrows. Depolarizing FRBs previously identified in the literature are overplotted as blue squares (repeaters; Y. Feng et al. 2022; A. Pandhi et al. 2026) and red circles (non-repeaters; P. A. Uttarkar et al. 2025). (Right) Rest-frame scattering timescale referenced to 600 MHz, where available, versus $\sigma_{\text{RM,host}}$ for the same sample of FRBs. Gold and blue shaded regions in both panels represent the 68% confidence interval around the median $|\text{RM}_{\text{host}}|$ and τ for repeaters, respectively. Uncertainties on $\sigma_{\text{RM,host}}$ are smaller than the markers. The test statistics and p-values from Pearson R tests in both cases are presented in the bottom right of each panel.

(i.e., their DM_{host} , RM_{host} , and $\langle B_{\parallel,\text{host}} \rangle$ are typical compared to the polarized FRB populations observed by CHIME/FRB and DSA-110; A. Pandhi et al. 2024; M. B. Sherman et al. 2024). In this section, we interpret the physical origins of the aforementioned spectropolarimetric propagation effects and discuss implications for population-level studies of FRB polarimetry.

6.1. Evidence for a magnetospheric emission process

The spectra of polarized radio emission of pulsars and FRBs are often analyzed for signatures of propagation effects through magnetized environments. Here, we focus on how the linear polarization fraction of FRB emission in our observed sample varies as a function of frequency, focusing on two physical scenarios: in this Section we discuss the power-law decrease in linear polarization towards higher frequencies, and in Section 6.2, we address the exponential depolarization of FRB emission at lower frequencies due to multipath propagation through inhomogeneous plasma.

As discussed in Section 4.1.3, a decrease in the L/I with increasing frequency is a commonly observed phenomenon in polarized pulsars that is typically attributed to the interaction of polarized orthogonal propagation modes with birefringent plasma in the neutron star mag-

netosphere. Despite the specific mechanism by which the resulting linearly polarized spectra is produced still being debated (e.g., M. A. Ruderman & P. G. Sutherland 1975; J. J. Barnard & J. Arons 1986; M. M. McKinnon 1997; A. von Hoensbroech et al. 1998; A. Karastergiou et al. 2005; S. Johnston et al. 2008), all of these models require the emission site of the pulses to be within the magnetosphere of the neutron star itself. While A. Pandhi et al. (2024) noted one FRB with a tentative increase in its L/I across 400 – 800 MHz, there have been no confirmed detections of this phenomenon in FRB signals until this work. In Section 5.2, we present three sources (FRBs 20230307A, 20230913A, and 20240229A) that show a decreasing L/I with increasing frequency, each depicting a $\sim 20 - 40\%$ lower L/I at the bottom of the DSA-110 frequency band compared to the top of the CHIME/FRB band. The L/I spectra of these FRBs are well fit by Equation 3, which includes a power-law increase of L/I towards lower frequencies motivated by the shape of polarized pulsar spectra in which this magnetospheric birefringence effect is observed (R. N. Manchester et al. 1975). In a radius-to-frequency mapping scheme, it is also expected that the burst width should increase with decreasing frequency (e.g., M. M. McKinnon 1997). However, due to

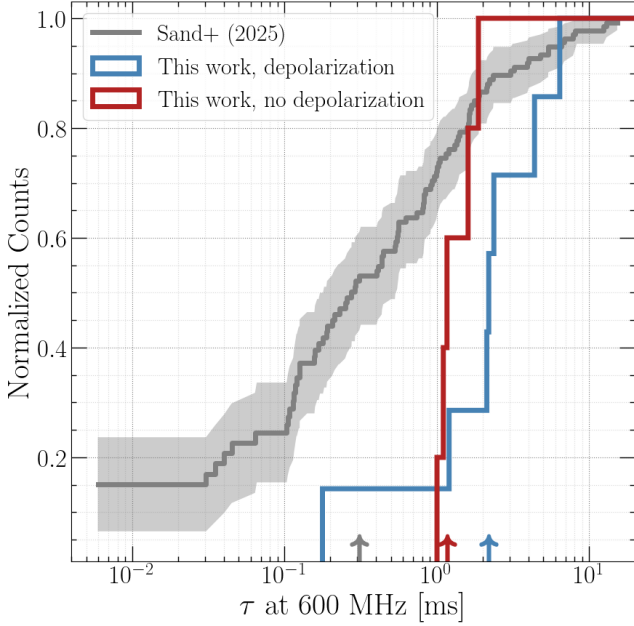


Figure 5. Cumulative distribution function of the scattering timescale (observer frame and referenced to 600 MHz) of the seven depolarizing (blue) and five non-depolarizing (red) codetected FRBs. The median of each distribution is marked by an upward-facing arrow in their respective colors. Overplotted in gray is the empirical cumulative distribution function of scattering timescales for the 137 FRBs from the first CHIME/FRB baseband catalog (K. R. Sand et al. 2025). For this data, lower limits have been incorporated as censored data and the shaded region represents the 95% confidence interval.

the significant scatter broadening at 400 – 800 MHz in conjunction with our lack of spectral coverage between 800 – 1280 MHz, it is challenging for us to unambiguously measure the burst width versus frequency in our FRB sample. Regardless, the remarkable resemblance between these polarized FRB spectra and pulsar observations likely point to a magnetospheric emission process for these FRBs, but below we explore a few other possible propagation effects that can produce confounding complexity in the L/I spectra.

Early spectropolarimetric observations of nearby galaxies revealed regions in some galaxies in which the polarized intensity fraction increased at longer wavelengths (e.g., in L-band observations of NGC 6946 and M51, respectively; R. Beck 1991; C. Horellou et al. 1992). Modeling by D. D. Sokoloff et al. (1998), and later expanded upon by C. Horellou & A. Fletcher (2014), showed that this polarized intensity spectra can be produced by differential Faraday rotation in a magnetic field configuration that is twisted along the LOS (e.g., helical) and that is colocated with the radio-emitting plasma. However, this explanation requires a

physically extended region along the LOS that is both emitting and inducing Faraday rotation of the observed signal, and that hosts a large-scale (with comparable correlation scale to the physical size of the emitting region) twisted magnetic field. These criteria are suitable for galaxies that host regular magnetic fields and emit synchrotron emission over $\sim$ kpc scales. However, these assumptions do not map on to FRB emission, which likely originates from a much more compact emission region that is effectively a point source along the LOS to the observer. Thus, it is infeasible for differential Faraday rotation in the presence of twisted magnetic fields to explain the observed L/I decrease at higher frequencies in FRBs 20230307A, 20230913A, and 20240229A.

Radio emission propagating through plasma with elliptically or linearly polarized natural wave modes will undergo some level of Faraday conversion, which can result in frequency-dependent conversion between linearly and circularly polarized intensity (V. N. Sazonov 1969; M. Kennett & D. Melrose 1998). This mechanism can arise under a variety of different physical conditions (e.g., D. B. Melrose 2010; A. Gruzinov & Y. Levin 2019; H. K. Vedantham & V. Ravi 2019; M. Lyutikov 2022) and, generally, the conversion between L/I and V/I becomes stronger at lower observing frequencies, but there is no uniform prescription for the spectral signature of this effect. Faraday conversion has been invoked to explain circular polarization of radio–mm emission from supermassive black holes, where the conversion can arise in extreme, magnetized plasma such as the hot accretion flows around the black hole (e.g., in Sgr A*; L. Huang & R. V. Shcherbakov 2011), or in their jets (J. F. C. Wardle & D. C. Homan 2003; N. R. MacDonald & A. P. Marscher 2018). Faraday conversion is also speculated to attenuate the polarization levels of some pulsar systems that show substantial circular polarization; here, the conversion would originate in highly magnetized plasma in the neutron star magnetosphere (R. T. Edwards & B. W. Stappers 2004; C. D. Ilie et al. 2019; M. E. Lower et al. 2024) or in the circumspace environment of a pulsar binary system (D. Li et al. 2023). Without well-calibrated circular polarization data at 400 – 800 MHz, direct tests of Faraday conversion models are challenging. Under the conservative assumption that Faraday conversion is frequency dependent such that conversion between L/I and V/I is stronger at lower frequencies, the expected signature in the L/I spectra would be a higher level of variation with decreasing frequency (e.g., L/I decreases monotonically or oscillates more rapidly at low ν). The observed L/I spectra of FRBs 20230307A, 20230913A, and 20240229A all rise towards lower frequencies, with

L/I across 400 – 800 MHz consistent with constant linear polarization or decrease that is well described by depolarization due to multipath propagation. Thus, our observations do not qualitatively match the expectations from Faraday conversion. Though our lack of well-characterized V/I spectra at 400 – 800 MHz and gap in observing frequencies between 800 – 1280 MHz makes it impossible to entirely rule out that the L/I spectra of these FRBs is shaped by Faraday conversion, we deem this scenario unlikely.

After evaluating various physical scenarios that can produce complex L/I spectra in FRB observations, we determine that the increase in L/I with decreasing frequency, seen in three of our FRB sources, is most likely due to the interaction of polarized orthogonal propagation modes with birefringent plasma in the magnetosphere of the FRB engine. Hence, emission sites for FRBs 20230307A, 20230913A, and 20240229A are likely in the magnetosphere of highly magnetized neutron stars, with higher frequency emission originating from deeper in the neutron star magnetosphere. This radius-to-frequency mapping has also been independently suggested as a potential origin for downward-drifting burst morphology (i.e., decreasing emission frequency over the burst duration) that is commonly seen in repeating FRBs (e.g., J. W. T. Hessels et al. 2019; M. Lyutikov 2020; H. Tong et al. 2022). Our codetected sample is comprised of apparently non-repeating, broadband FRBs, suggesting that some FRBs may share common emission sites despite their visually distinct burst morphologies. In all, our spectropolarimetric analyses highlight an unexplored avenue in wideband FRB surveys to investigate the origins of FRB emission on a population level.

6.2. Prevalence of depolarizing environments

Measurements of spectral depolarization in FRB sources have thus far been exceedingly rare (9 in total), with most detections being associated with repeating FRBs that have long activity periods, allowing for follow-up observations at multiple frequency bands (Y. Feng et al. 2022; A. Pandhi et al. 2026). There is only one concrete depolarization measurement (P. A. Uttarkar et al. 2025), and one tentative detection (A. Pandhi et al. 2024), among the non-repeating FRB population. Non-repeating FRBs still show a typically high median $L/I \sim 0.65$ at 400 – 800 MHz which is consistent with the distribution of non-repeater L/I at 1400 MHz (A. Pandhi et al. 2024), pointing to the lack of strong, population-level depolarization. These results have led to the preferred association of inhomogeneous, depolarizing environments with repeating FRBs embedded in

extreme magnetoionic plasma (Y.-P. Yang et al. 2020, 2022; Y.-P. Yang 2026).

However, the bandwidth of non-repeating FRBs is limited to the detecting telescope’s frequency coverage (which, in the case of CHIME/FRB and DSA-110, is smaller than the emitting bandwidth of most non-repeating FRBs), narrowing the range of depolarizing environments that the observations are sensitive to. Thus, it is possible that non-repeating FRBs also experience spectral depolarization, though to a lesser extent than some of the extreme repeating FRBs, such as FRB 20121102A. Despite having DM_{host} , DM_{host} , and $\langle B_{\parallel, \text{host}} \rangle$ values typical of the non-repeating FRB population (A. Pandhi et al. 2024; M. B. Sherman et al. 2024), we discover measurable depolarization in 7/12 of our sources, nearly doubling the number of FRB sources in which depolarization has been detected. The $\sigma_{\text{RM}, \text{host}}$ of the non-repeating FRBs in this work are lower than those typically found in repeating FRBs, providing support to the hypothesis that repeating FRBs originate from more inhomogeneous magnetoionic environments than non-repeaters, though a larger population study that also incorporates non-detections is needed to further test this claim. Overall, our results suggest that depolarization in the FRB population, particularly in non-repeating FRBs, is more common than previously expected, and low-frequency FRB surveys are vital for further constraining population-level depolarization.

Depolarization due to multipath propagation, as proposed in Equation 2 (B. J. Burn 1966), is seldom invoked to explain polarized spectra of radio pulsars, as a precipitous decrease in their L/I with decreasing frequency is not commonly observed. M. Xue et al. (2019) note that the L/I of PSR J0742–2822 decreases dramatically below ~ 300 MHz, finding a best-fit $\sigma_{\text{RM}} = 0.13 \text{ rad m}^{-2}$ (taken to be in the rest frame as the source is Galactic), and attributing this depolarization to an intervening inhomogeneous screen in the Gum nebula. C. Sobey et al. (2021) also note that the complete depolarization in the L/I spectrum of PSR J1721–3532 at $\lesssim 1200$ MHz, with $\sigma_{\text{RM}} \sim 20 \text{ rad m}^{-2}$ is likely due to an H II region along the LOS. In a study focusing on $\lesssim 200$ MHz observations of 16 pulsars, A. Noutsos et al. (2015) see a drop in L/I at the $\sim 10\%$ level for PSRs B0136+57, B1508+55, and B2111+46. Though they do not explicitly fit the data with a model similar to Equation 2, the large L/I decrease and low frequencies would result in order $\sigma_{\text{RM}} \lesssim 10^{-1} \text{ rad m}^{-2}$. P. F. Wang et al. (2023) conduct a systematic polarimetric analysis of 682 pulsars between 1000 – 1500 MHz, finding that the vast majority do not show frequency-dependent L/I evolution; only PSRs J0358+5413, J1857+0212, and

J2013+3845 show a $\gtrsim$ few percent decrease in L/I with decreasing frequency. The lack of spectral depolarization in most polarized pulsars even down to low frequencies ($\lesssim 400$ MHz), in conjunction with the significant depolarization in PSRs J0742–2822 and J1721–3532 likely being due to intervening, dense Galactic structures, may suggest that most pulsars are not enshrouded in highly inhomogeneous plasma. Meanwhile, at least some FRBs are embedded in extremely inhomogeneous plasma (e.g., FRBs 20121102A and 20190520B with $\sigma_{\text{RM,host}} \sim 44 \text{ rad m}^{-2}$ and $\sigma_{\text{RM,host}} \sim 340 \text{ rad m}^{-2}$, respectively; Y. Feng et al. 2022), and we see significant depolarization at 400 – 1530 MHz in more than half of our FRB sample. Therefore, FRBs, on average, may originate from magnetoionic environments that are more inhomogeneous (and thus potentially younger and more compact) than those surrounding or intersecting pulsars in the Milky Way, though a more complete spectropolarimetric accounting of both FRB and pulsar populations at comparable frequencies is required to rigorously test this hypothesis.

We also find the first statistically significant ($\geq 3\sigma$) detection of a correlation between $\sigma_{\text{RM,host}}$ and $|\text{RM}_{\text{host}}|$, and a marginally significant ($\sim 2.8\sigma$) correlation between $\sigma_{\text{RM,host}}$ and τ_{host} . In addition, we see some evidence that the scattering timescale of depolarizing FRBs may be larger than FRBs that are not undergoing depolarization (see Figure 5). The $\sigma_{\text{RM,host}} - |\text{RM}_{\text{host}}|$ correlation confirms that most of the Faraday rotation attributed to the FRB host galaxy is contributed by the same local, inhomogeneous environment that produces depolarization of the radio signal. A larger sample of FRBs with detailed spectropolarimetric analysis will further test these correlations, particularly between $\sigma_{\text{RM,host}} - \tau_{\text{host}}$, which remains only marginally significant with the current population of depolarizing FRBs. If indeed higher levels of scattering are indicative of depolarization, we note that $\sim 13\%$ of FRBs from the first CHIME/FRB baseband catalog have scattering timescales in excess of the mean observed τ of our depolarizing FRBs (K. R. Sand et al. 2025), making them potential candidates for further spectropolarimetric analyses.

6.3. Limitations due to irregular frequency coverage

Here, we discuss the limitations on L/I spectra modeling imposed by our unique, cross-facility data products. The gap in frequency coverage between the bottom of the DSA-110 band (1280 MHz) and the top of the CHIME/FRB band (800 MHz) is the foremost impediment for this analysis insofar as it restricts us from irrefutably showing that the codetected FRBs are

the same emission components extending from 400 – 1530 MHz (e.g., the component detected by DSA-110 only extends to $800 \text{ MHz} \leq \nu \leq 1280 \text{ MHz}$ and the CHIME/FRB detection is of a separate component whose emitting band only starts at $\leq 1280 \text{ MHz}$). In this context, H. Cho et al. (2020) and A. Pandhi et al. (2024) showed that different subcomponents of the same burst can have distinct polarization properties, which could complicate the L/I modeling that we perform. This is only a concern for FRBs that do not show consistent L/I values between the DSA-110 band and, at least, the top of the CHIME/FRB band: the three FRBs best fit by Equation 3 and FRB 20220506D (which has too low S/N to measure a robust L/I at 400 – 800 MHz). Given the limitations of our data, we still argue that the FRBs in our sample are, most likely, detections of the same emission components across 400 – 1530 MHz for a number of reasons: (i) they have consistent sky positions, DMs, RMs, and arrival times between the two telescopes, (ii) all bursts are broadband in the two frequency ranges (400 – 800 MHz and 1280 – 1530 MHz, respectively), and (iii) the number of components in the FRB burst envelopes are generally in agreement in the two data sets (this argument is described in more detail in paper I). We also measure the L/I spectra averaged across all detected components in time, which suppresses any spurious frequency-dependent L/I structure that may be introduced by intraburst polarization variation. Further, this gap in our frequency coverage between 800 – 1280 MHz leads to the degeneracy between ν_c and α seen in the model-fitting results for FRBs 20230307A, 20230913A, and 20240229A.

6.4. Implications of PA flattening from scattering for the FRB population

The PA profiles across the duration of a radio transient is a key observable that helps to constrain its emission mechanism. Specifically, they serve as a diagnostic tool to distinguish between models in which FRB emission originates near the surface of a compact object (“close-in” models, e.g. from coherent-curvature radiation or magnetic reconnection; P. Kumar et al. 2017; Y.-P. Yang & B. Zhang 2018; M. Lyutikov & S. Popov 2020; M. Lyutikov 2021) and those where the emission is produced at large distances from the FRB engine (“far-out” models, e.g. by synchrotron maser emission; B. D. Metzger et al. 2019; A. M. Beloborodov 2020). In the former case, the viewing geometry to a small emission region corotating with a rapidly-spinning compact object can cause the observed PA to vary on $\lesssim$ ms timescales; however, in the latter case, our sightline to the FRB-

emitting region is not expected to significantly change over such short timescales.

A geometric model describing the projection of beamed emission from a dipolar neutron star magnetic field towards an observer (i.e., the “rotating vector model”; V. Radhakrishnan & D. J. Cooke 1969) has proved to be well-suited in explaining the rich phenomenology of PA variability observed in most radio pulsars (e.g., S. Johnston et al. 2023). While some FRBs show evidence for PA evolution that is in agreement with the rotating vector model, hence preferring a close-in emission mechanism (R. Mckinven et al. 2025), the majority of FRBs display little to no measurable PA changes across their burst durations. Previous polarimetric surveys with CHIME/FRB and DSA-110 have found that $\sim 80-90\%$ of polarized FRBs depict PA profiles consistent with no variability (A. Pandhi et al. 2024; M. B. Sherman et al. 2024). Such a high occurrence rate of constant PA profiles are challenging to reconcile with a scenario in which all FRBs are produced by close-in emission mechanisms with diverse and variable viewing geometries; thus, these results appear to indicate that at least a large fraction of the FRB population may be consistent with far-out emission models.

However, the above conclusion presupposes that there are no confounding systematic biases that may lead to an over representation of flat PA profiles in these surveys. Such an effect has been noted for radio pulsars, where scatter broadening of the pulse leads to the mixing of signals from different pulse phases, averaging over the temporal evolution of the PA profile and suppressing PA variability (e.g., X. H. Li & J. L. Han 2003; P. F. Wang et al. 2023; W. C. Jing et al. 2025). There are two main impediments for constraining this PA flattening effect in FRBs: (i) the vast majority of FRB sources ($\sim 97\%$) are non-repeaters, so current FRB surveys do not have the necessary low-frequency, broadband coverage and large field of view required to measure this PA flattening in a large number of FRBs, and (ii) repeating FRBs, which may be observed across multiple frequency bands, ubiquitously have narrowband emission, and they often show burst-to-burst variability in their polarization properties, making it challenging to definitively observe this effect in a given burst. PA flattening in FRBs has only been reported and discussed for FRB 20250316A (CHIME/FRB Collaboration et al. 2025), where the serendipitous combination of extremely high S/N , intrinsic PA variability, and scattering timescale (such that the the burst profile only began to become dominated by scatter broadening at $\lesssim 600$ MHz), made detection possible.

In Section 5.3, we show that 2/12 codetected FRBs exhibit PA flattening between 1530 MHz and 400 MHz. While our sample size is not large enough to place strong constraints on the rate at which intrinsically variable PA profiles are suppressed at low frequencies, multiple detections of PA flattening in our limited sample does suggest that the effect serves as a significant systematic bias that must be addressed in polarimetric FRB studies. Importantly, many of the DSA-110 polarized FRB sample that do not show significant scatter broadening above 1280 MHz, presented here as codetected FRBs or previously by M. B. Sherman et al. (2024), still have flat PA profiles. This is in agreement with observations at observing frequencies $\gtrsim 1400$ MHz of some repeating FRBs, in which many bursts show flat PA profiles despite the lack of scatter broadening (e.g., D. Michilli et al. 2018; K. Nimmo et al. 2021; G. H. Hilmarsson et al. 2021; X. Liu et al. 2025; T.-C. Wang et al. 2026). Overall, these observations imply that many FRBs do indeed have intrinsically flat PA profiles (i.e., our viewing geometry of the FRB emission is not changing on timescales of the bursts themselves), but that a fraction of bursts showing flat PA profiles are contaminants from PA flattening due to scatter broadening, which is more prominent at lower frequencies. Thus, future FRB polarization studies should be mindful of, and attempt to account for, this systematic bias by analyzing PA variability in FRBs whose burst profiles are minimally modulated by scatter broadening.

7. CONCLUSIONS

We report on the spectropolarimetric properties of 12 FRBs codetected by CHIME/FRB and DSA-110. The joint spectral coverage of these two telescopes, spanning two octaves in frequency (400–1530 MHz), enables tests of subtle propagation effects imprinted in the polarized spectra of these FRBs. We test the data against three physically motivated models of the frequency evolution of the linear polarization fraction, L/I : (i) a constant L/I versus frequency, (ii) spectral depolarization in which the L/I decreases exponentially at lower frequencies, and (iii) model (ii) convolved with a decrease at high frequencies from propagation through a birefringent magnetosphere. We also analyze the evolution of polarization angle, PA, with frequency, focusing on how scattering of the FRB emission may effect the observed PA profile at different frequencies. Our key conclusions from this work are:

1. **Magnetospheric FRB emission process.** In three sources (FRBs 20230307A, 20230913A, and 20240229A), we discover a significant increase in L/I with decreasing frequency between 800 –

1530 MHz. The polarized spectra of these FRBs are consistent with the interaction of polarized, orthogonal propagation modes with birefringent plasma in the magnetosphere of a neutron star, a phenomenon that has been previously observed in pulsar observations. At higher frequencies, the angular separation between the two modes becomes smaller, causing their emission to become mixed, and leading to decreased L/I . We interpret these results as the emission sites of FRBs 20230307A, 20230913A, and 20240229A being in the magnetosphere of highly magnetized neutron stars.

2. Inhomogeneous FRB environments. We measure spectral depolarization in 7/12 of the codetected FRBs, nearly doubling the number of depolarized FRB sources currently known, and characterizing the first sample of such sources among non-repeating FRBs. Our results indicate that depolarization is likely more prevalent in the FRB population than previously expected, but that the exponential drop in L/I is only be observable for many sources at low frequencies $\lesssim 600$ MHz. These findings suggest that a significant fraction of FRBs may originate from inhomogeneous local environments.

3. Systematic bias in PA studies. We find scattering-induced flattening in the PA profiles of two FRBs from 1280 – 1530 MHz, where they show PA variation of $\gtrsim 50^\circ$, to 400 – 800 MHz, where their PA profiles are consistent with being constant. These results provide only the second and third such cases in which PA profile flattening has been clearly measured in FRB spectra. Hence, FRB polarization studies, particularly at low frequencies where scatter broadening is amplified, are biased towards detecting preferentially constant PA profiles. We argue that a sample of FRB sources observed at high frequencies, or one at low frequencies that is controlled for scattering, are necessary to determine the fraction of FRBs that have intrinsically constant PA profiles.

This work acts a blueprint for the range of science that will be possible with broadband FRB surveys on upcoming radio telescopes, namely the Canadian Hydrogen Observatory and Radio transient Detector (CHORD; K. Vanderlinde et al. 2019) and the full Deep Synoptic Array (DSA; G. Hallinan et al. 2019). Both of these instruments will have wide bandwidths centered at low frequencies (300 – 1500 MHz for CHORD and 700 – 2000 MHz for DSA) that are ideal for constraining the spectropolarimetric propagation effects highlighted

here. They will also bypass the largest limitation of this work: the data gap between 800 – 1280 MHz, enabling a more complete modeling of FRB spectra across these frequencies. The increased sensitivity of CHORD and DSA will also drastically increase the population of detected FRBs, as they are projected to discovered on the order of 10^4 FRBs in their first few years of operations; they will empower analyses similar to this work to be conducted on a much larger scale, thus constraining the origins of FRB emission mechanisms and the properties of their immediate environments on a population level.

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